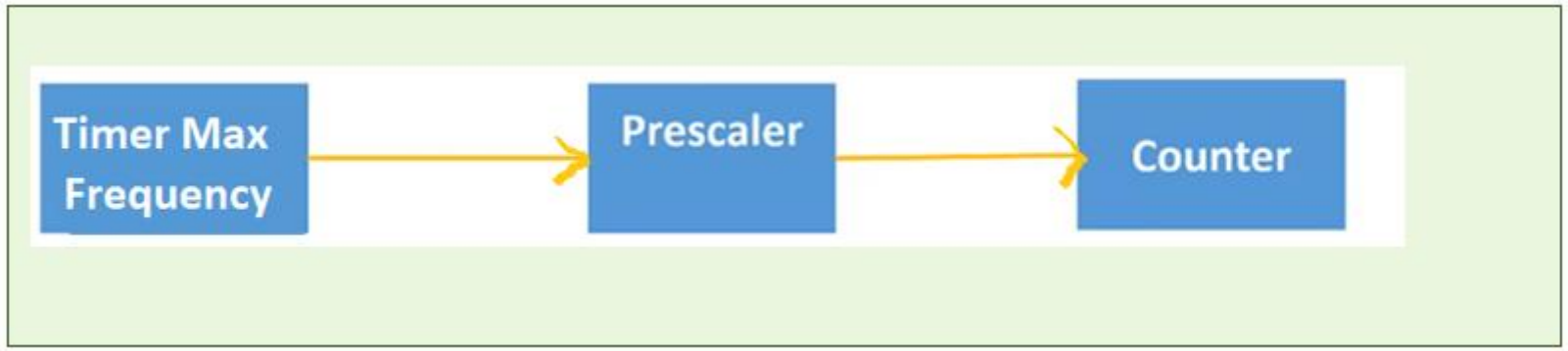
Choosing Prescalar and Period for timer

- **Timer:**
  - **Solution 3 (Use Prescaler):**



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Advanced Control	TIM1, TIM8	16-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	Yes	APB2 Timer Clock	168
General Purpose	TIM2, TIM5	32-bit	Up, Down, Up/down	between 1 and 65536	Yes	4	No	APB1 Timer Clock	84
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- **TIMx Counter** (TIMx_CNT, address offset 0x24)
  - 16-bit binary counter (32 in TIM2, TIM5)

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- **TIMx Prescale Register** (TIMx_PSC, address offset 0x28)
  - Clock prescale value (16 bits)

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$2^{16} \rightarrow$ 65,536

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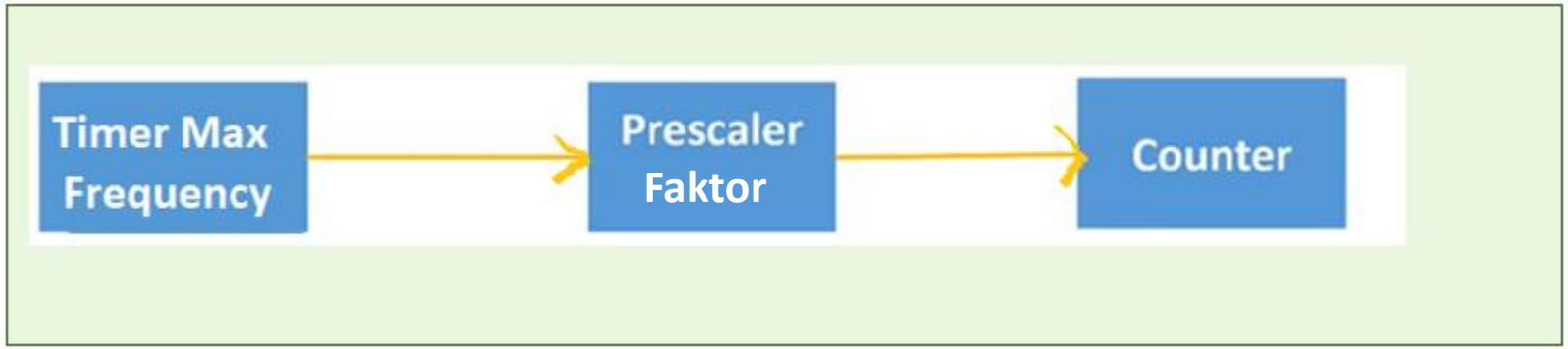

Prescalar Factor = 2^x

$x \longrightarrow$ Prescalar Value

```
TIM_TimeBaseStructure.TIM_Prescaler =
```

- Modify the blinking frequency in Timer_Template.cpp file to 0.5Hz

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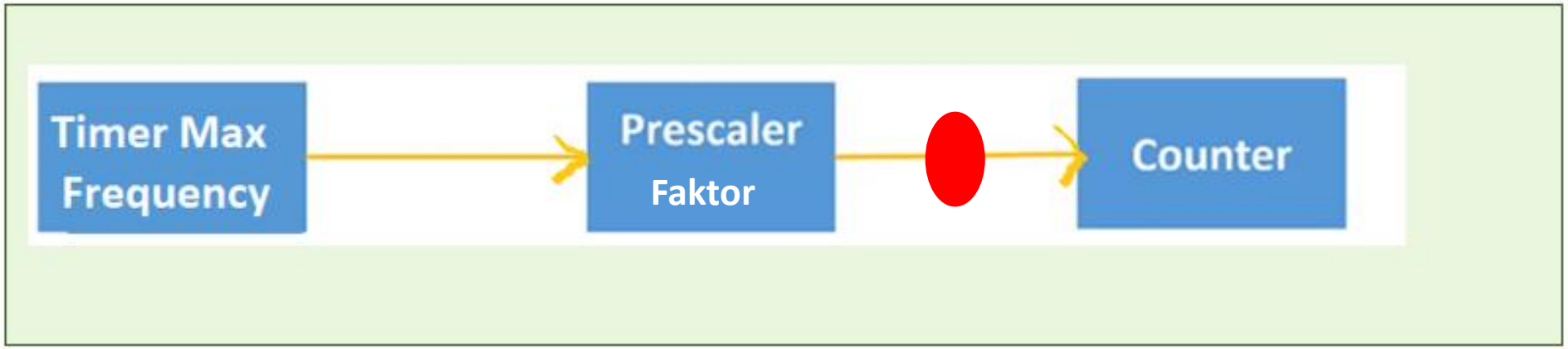

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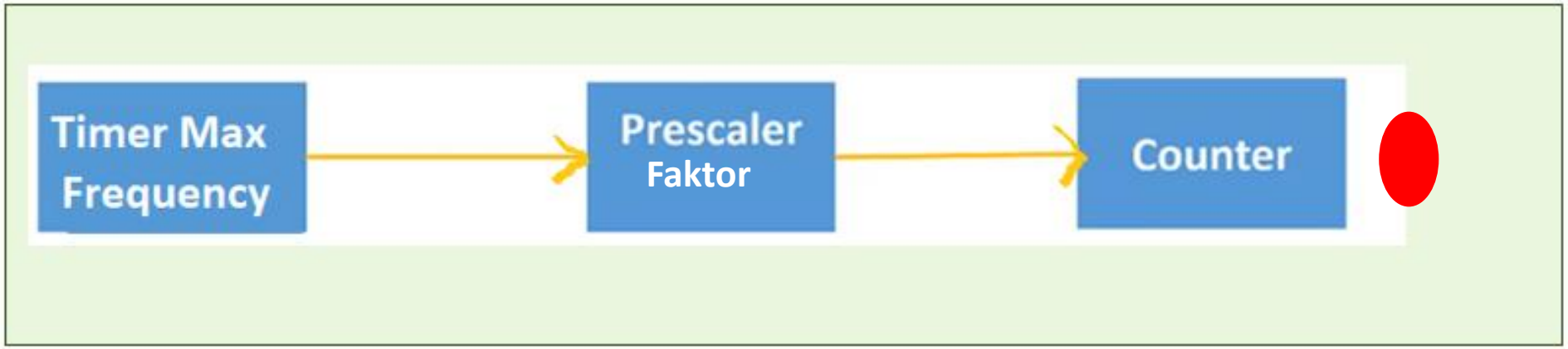

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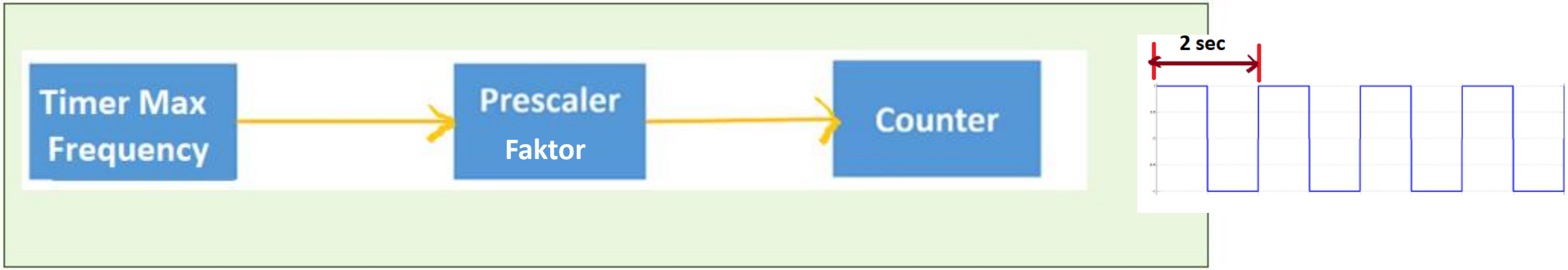

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$$\boxed{Prescaler = \frac{Timer_Max_Frequency}{Frequency_After_Prescaler} - 1}$$

Value

$$\boxed{Period = \frac{Frequency_After_Prescaler}{Timer_Desired_Frequency} - 1}$$

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$$Prescaler = \frac{Timer_Max_Frequency}{Frequency_After_Prescaler} - 1$$

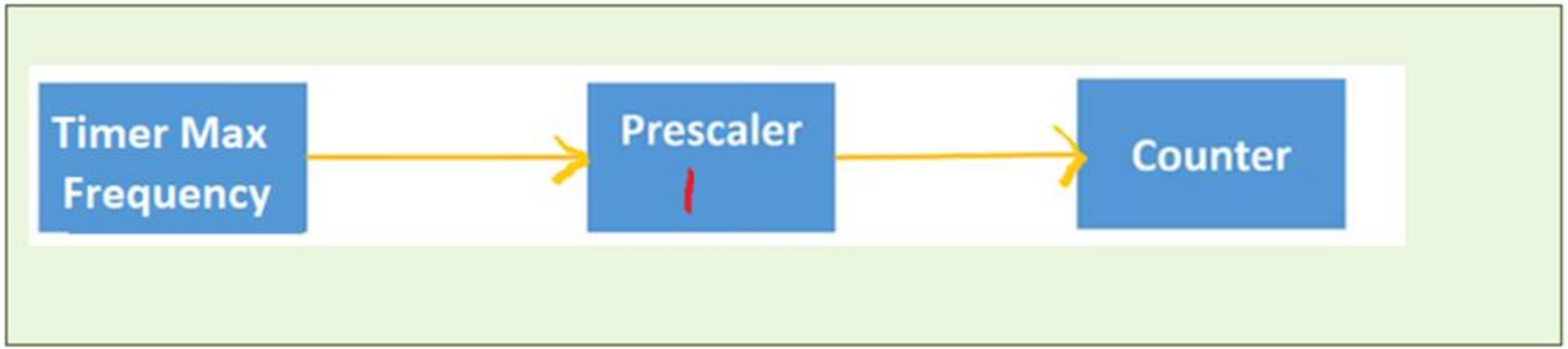
$$\frac{84\ MHz}{84\ MHz} - 1 = 0$$

So

- Prescalar Value = 0 (This will be written in the code)

  Prescalar Factor = 1 (It should be used to make sure That it is under allowable region As given in the table on next slide Where 1 is included.)

$$Period = \frac{Frequency_After_Prescaler}{Timer_Desired_Frequency} - 1$$



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$$\text{Period} = \frac{84\ \text{MHz}}{0.5\ \text{Hz}} - 1$$

Period = 167 999 999

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The Counter can count from 0 to **65535**

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The Counter can count upto

2 ^ 32 =

4,294,967,296

```
void Setup_Timer(void)
{
  /* TIM2 clock enable */
  RCC_APB1PeriphClockCmd(RCC_APB1Periph_TIM2, ENABLE);
            .
            .
            .
            .
            .
            .
            .

  TIM_TimeBaseInit(TIM2, &TIM_TimeBaseStructure);
  /* TIM2 enable counter */
  TIM_Cmd(TIM2, ENABLE);


}
```